MLEARN · Tema 2 · Decision theory

Duda, Hart & Stork, caps. 2–3 · diap. 1–85

1 · BAYES Y MAP (d. 6–16)

Bayes (d. 7)	$\Pr(\omega_j\|\mathbf{x}) = \frac{f(\mathbf{x}\|\omega_j)\Pr(\omega_j)}{f(\mathbf{x})}$, $f(\mathbf{x}) = \sum_i f(\mathbf{x}\|\omega_i)\Pr(\omega_i)$ · posterior = verosimilitud × prior / evidencia
MAP (d. 13–15)	$\hat{\omega} = \arg\max_i f(\mathbf{x}\|\omega_i)\Pr(\omega_i)$ · minimiza $\Pr(e)$ · $\Pr(e\|\mathbf{x}) = 1 - \max_i \Pr(\omega_i\|\mathbf{x})$ · priors iguales → ML
Error, 2 clases (d. 17)	$\Pr(e) = \Pr(\omega_1)\int_{\mathcal{R}_2} f_1\,d\mathbf{x} + \Pr(\omega_2)\int_{\mathcal{R}_1} f_2\,d\mathbf{x}$

2 · RIESGO MÍNIMO (MBR) (d. 17–22)

Riesgo	$\mathcal{R} = \sum_{i,j}\pi_{ij}\Pr(\omega_i\|\omega_j)\Pr(\omega_j)$ · π_{ij}: decidir ω_i siendo ω_j · $\Pr(\omega_i\|\omega_j) = \int_{\mathcal{R}_i} f(\mathbf{x}\|\omega_j)\,d\mathbf{x}$
Regla	$\mathcal{C}(\omega_i\|\mathbf{x}) = \sum_j \pi_{ij}\Pr(\omega_j\|\mathbf{x})$, $\hat{\omega} = \arg\min_i \mathcal{C}(\omega_i\|\mathbf{x})$ · costes 0-1 → MAP
Binaria (d. 20)	$\frac{\Pr(\omega_1\|\mathbf{x})}{\Pr(\omega_2\|\mathbf{x})} \underset{\omega_2}{\overset{\omega_1}{\gtrless}} \gamma = \frac{\pi_{12}-\pi_{22}}{\pi_{21}-\pi_{11}} \iff \Pr(\omega_1\|\mathbf{x}) \gtrless \frac{\gamma}{1+\gamma}$ · cociente de *posteriores*

3 · NEYMAN-PEARSON Y MINIMAX (d. 23–28)

Probabilidades	$P_D = \int_{\mathcal{R}_2} f(\mathbf{x}\|\omega_2)\,d\mathbf{x}$, $P_{FA} = \int_{\mathcal{R}_2} f(\mathbf{x}\|\omega_1)\,d\mathbf{x}$
NP (d. 25)	$\max P_D$ con $P_{FA} \le \alpha$: $\frac{f(\mathbf{x}\|\omega_2)}{f(\mathbf{x}\|\omega_1)} \underset{\omega_1}{\overset{\omega_2}{\gtrless}} \gamma$ · Lagrangiano $\int_{\mathcal{R}_2}(f_2 - \gamma f_1)\,d\mathbf{x} + \gamma\alpha$ · sin priors
1D gaussiano	$\mathcal{N}(0,1)$ frente a $\mathcal{N}(D',1)$: $\gamma' = Q^{-1}(\alpha)$, $P_D = Q(\gamma' - D')$ · $\alpha = 0{,}05,\ D' = 2 \Rightarrow P_D = 0{,}639$
Minimax (d. 27)	$(\pi_{11}-\pi_{22}) + (\pi_{21}-\pi_{11})\int_{\mathcal{R}_2} f_1 - (\pi_{12}-\pi_{22})\int_{\mathcal{R}_1} f_2 = 0$ · costes 0-1: $P_{FA} = P_{\text{miss}}$

4 · GAUSSIANAS Y DISCRIMINANTES (d. 31–43)

Discriminantes	MBR: $-\mathcal{C}(\omega_i\|\mathbf{x})$ · MAP: $\Pr(\omega_i\|\mathbf{x})$ · NP: $f(\mathbf{x}\|\omega_i)$ · decidir el mayor; vale cualquier función creciente
Gaussiana (d. 33)	$f(\mathbf{x}) = \frac{1}{(2\pi)^{d/2}\|\mathbf{C}\|^{1/2}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T\mathbf{C}^{-1}(\mathbf{x}-\boldsymbol{\mu})}$ · $\Pr(x > \mu + 2\sigma) = Q(2) = 2{,}28\,\%$
Lineal (d. 34–35)	$\mathbf{z} = \mathbf{A}^T\mathbf{x}$: $\boldsymbol{\mu}_z = \mathbf{A}^T\boldsymbol{\mu}$, $\mathbf{C}_z = \mathbf{A}^T\mathbf{C}\mathbf{A}$ · blanqueado $\mathbf{A} = \mathbf{U}\boldsymbol{\Lambda}^{-1/2}$, $\boldsymbol{\Lambda}^{-1/2} = \operatorname{diag}(1/\sqrt{\lambda_i})$
QDA (d. 39)	$g_i = -\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_i)^T\mathbf{C}_i^{-1}(\mathbf{x}-\boldsymbol{\mu}_i) - \frac{1}{2}\ln\|\mathbf{C}_i\| + \ln\Pr(\omega_i)$
Frontera (d. 40–41)	$\mathbf{x}^T\mathbf{A}\mathbf{x} + \mathbf{v}^T\mathbf{x} + e = 0$, $\mathbf{A} = \frac{1}{2}(\mathbf{C}_j^{-1} - \mathbf{C}_i^{-1})$ · autovalores de $\mathbf{A}$: mismo signo → elipse; opuesto → hipérbola; uno nulo → parábola; $\mathbf{A} = \mathbf{0}$ → recta

5 · CASOS PARTICULARES (d. 44–50)

LDA ($\mathbf{C}$ común) (d. 48)	$\mathbf{w}^T(\mathbf{x}-\mathbf{x}_0) = 0,\ \mathbf{w} = \mathbf{C}^{-1}(\boldsymbol{\mu}_i - \boldsymbol{\mu}_j),\ \mathbf{x}_0 = \frac{1}{2}(\boldsymbol{\mu}_i + \boldsymbol{\mu}_j) - \frac{\ln[\Pr(\omega_i)/\Pr(\omega_j)]}{(\boldsymbol{\mu}_i-\boldsymbol{\mu}_j)^T\mathbf{C}^{-1}(\boldsymbol{\mu}_i-\boldsymbol{\mu}_j)}(\boldsymbol{\mu}_i - \boldsymbol{\mu}_j)$
MBR con $\mathbf{C}$ común	igual $\mathbf{w}$; en $\mathbf{x}_0$, $\Pr(\omega_i) \to (\pi_{ji}-\pi_{ii})\Pr(\omega_i)$ y $\Pr(\omega_j) \to (\pi_{ij}-\pi_{jj})\Pr(\omega_j)$ · 3 clases con costes generales: frontera curva
Diagonal / isotrópica (d. 47)	$g_i = \ln\Pr(\omega_i) - \frac{1}{2}\sum_j \frac{(x_j-\mu_{ij})^2}{\sigma_j^2}$ · $\mathbf{C} = \sigma^2\mathbf{I}$ y priors iguales: media euclídea más cercana (mediatriz)
Binarias (d. 50)	$h(\mathbf{x}) = \sum_i x_i \ln\frac{p_i(1-q_i)}{q_i(1-p_i)} + \sum_i \ln\frac{1-p_i}{1-q_i} + \ln\frac{\Pr(\omega_1)}{\Pr(\omega_2)}$ (lineal) · $p_i = 1 - q_i = p$: mayoría, $P_e = \sum_{k\le(d-1)/2}\binom{d}{k}p^k(1-p)^{d-k}$

6 · ESTIMACIÓN ML Y POCOS DATOS (d. 51–65)

ML (d. 54)	$\hat{\boldsymbol{\theta}}_i = \arg\max \sum_k \ln f(\mathbf{x}_{k,i}\|\boldsymbol{\theta}_i)$ · asintóticamente insesgado y eficiente
Estimadores (d. 59–60)	$\hat{\boldsymbol{\mu}} = \frac{1}{N}\sum \mathbf{x}_k$, $\hat{\mathbf{C}} = \frac{1}{N}\sum(\mathbf{x}_k - \hat{\boldsymbol{\mu}})(\mathbf{x}_k - \hat{\boldsymbol{\mu}})^T$ ($E\{\hat{\sigma}^2\} = \frac{N-1}{N}\sigma^2$) · $\hat{p}_k = \frac{1}{N}\sum_j x_{k,j}$ · priors N_k/N
LDA / QDA (d. 61)	LDA: $\hat{\mathbf{C}} = \frac{1}{N}\sum_k N_k\hat{\mathbf{C}}_k$ · QDA: un $\hat{\mathbf{C}}_k$ por clase · singular si $N_k \leq d$
RDA (d. 63)	$\hat{\mathbf{C}}_k(\alpha) = \alpha\hat{\mathbf{C}}_k + (1-\alpha)\hat{\mathbf{C}}$ (1 = QDA, 0 = LDA) · $(1-\gamma)\hat{\mathbf{C}}_k(\alpha) + \gamma\hat{\sigma}^2\mathbf{I}$, $\hat{\sigma}^2 = \frac{1}{d}\,\mathrm{tr}$
Naive Bayes (d. 64–65)	$f(\mathbf{x}\|\omega) = \prod_j f(x_j\|\omega)$ · Bernoulli → lineal · gaussiano → QDA diagonal (d parámetros frente a $\frac{d(d+1)}{2}$)

7 · INDICADORES (d. 67–84)

Cuatro probabilidades	P_D (sensibilidad, recall) · P_{FA} · $P_{\text{miss}} = 1 - P_D$ · $P_{CR} = 1 - P_{FA}$ (especificidad)
ROC (d. 71)	$D' = \|\mu_2 - \mu_1\|/\sigma$ · curva $(Q(\gamma'),\, Q(\gamma' - D'))$ · AUROC $= 1 - Q(D'/\sqrt{2})$: 0,760 / 0,921 / 0,983 para $D' = 1, 2, 3$
Métricas (d. 74)	$R = \frac{TP}{TP+FN}$, $S = \frac{TN}{TN+FP}$, $P = \frac{TP}{TP+FP}$, $A = \frac{TP+TN}{N}$, $F_1 = \frac{2PR}{P+R}$
Test médico (d. 79–80)	$\Pr(\mathrm{s}\|+) = \frac{\mathrm{se}\,p}{\mathrm{se}\,p+(1-\mathrm{sp})(1-p)}$ · 0,997 / 0,985 / 0,001 → 0,062 → (2.º +) 0,816
Confusión (d. 83)	TP 16, FN 30, FP 10, TN 144 → R 0,348 · S 0,935 · P 0,615 · A 0,80 · F1 0,444
Qué métrica (d. 84)	spam → precisión · enfermedad → recall · subir γ': ↑P, ↓R

8 · CONCLUSIÓN (d. 85)

Reglas	MBR: pdf + priors + costes · MAP: pdf + priors · NP: solo pdf (y α)
Modelos	LDA (lineal) · QDA (cuadrático) · Naive Bayes · no paramétricos (Parzen, KNN) · cualquiera se combina con cualquier regla